Pancreatic Cancer Prognosis Using Clinical and Radiomic Data

Machine Learning Theory Mini-Project | Preliminary executed baseline | Updated 11 October 2026

1. Problem Statement

This project investigates whether patient-level clinical information can help rank survival outcomes in pancreatic ductal adenocarcinoma (PDA). The provided Stanford CS229 reference paper motivates comparing clinical, radiomic, and combined features using survival analysis. The current executed experiment is a clinical-only baseline; radiomic features have not yet been integrated.

2. Dataset and Outcome Definition

The user-exported CPTAC PDA Discovery Study clinical table contained 166 records. Survival time was defined as Days to Death for records whose vital status was Dead, and Days to Last Follow Up for records whose status was Alive. The event indicator is 1 for observed death and 0 for alive/right-censored. Rows with missing or unreported status, missing duration, or non-positive duration were excluded. The resulting analysis cohort contains 143 cases: 115 observed deaths and 28 right-censored cases. Of these, 73 case identifiers also appear in the TCIA CPTAC-PDA imaging metadata manifest (110 unique imaging IDs); this ID overlap does not prove that radiomic features are available for all 73 cases.

3. Features and Method

A penalized Cox proportional hazards model was fitted using three baseline clinical predictors: age at diagnosis (source age in days converted to approximate years), sex (female indicator), and tumor grade grouped as high grade (G3/G4) versus other grades. Five-fold shuffled cross-validation was run with random seed 42. The feature set is deliberately small given the cohort size and the low counts in some grade categories. All imputation/scaling is fitted within each training fold.

4. Results

The mean fold concordance index (C-index) was 0.567 (fold standard deviation 0.043); the pooled out-of-fold C-index was 0.574. A C-index near 0.5 indicates ranking near chance, while higher values indicate better pairwise risk ranking. These values are exploratory internal estimates from a small cohort and limited predictors, not external validation. They should not be interpreted as clinical utility.

5. Limitations and Next Steps

The trained model uses clinical predictors only. No radiomic feature values were present in the supplied clinical export, and raw medical images/annotations have not yet been processed into a radiomic feature table. Therefore, this run does not reproduce the reference paper's clinical-plus-radiomic model or its reported C-index. The next milestone is to obtain tumor annotations and radiomic features for matched patient IDs, then compare clinical-only, radiomic-only, and combined models with leakage-aware cross-validation. This project is educational research, not a diagnostic or clinical decision tool.

References

Jamalian, A. (2020). Stanford CS229 project report: Pancreatic cancer prognosis using clinical and radiomic data (reference supplied for this assignment). CPTAC-PDA, The Cancer Imaging Archive: https://www.cancerimagingarchive.net/collection/cptac-pda/. MultiOmicPDA research repository: https://github.com/FraBer95/MultiOmicPDA.